Cover Page

Problem Statement/Concept Title	Smart Eco-Safe Dual-Chamber Sanitary Pad Incinerator for Schools
Theme	Health, Sanitation & Environment (SDG 3, SDG 9, SDG 13)
Team Name (Registered on portal)	[Enter your team name]
Team ID (Registered on portal)	[Enter your team ID]
State/UT	[Enter your State/UT]

Team Name

Problem – Proposed Solution

Core Problem

- No safe sanitary waste disposal in many rural schools: pads dumped in toilets or burnt openly
- Existing incinerators: no temperature control, incomplete burning, smoke and odour
- CO/VOC emissions, no monitoring, no child-safety locks
- Leads to infections, stigma and girls' absenteeism

Proposed Solution

- Dual-chamber incinerator: primary burn + secondary re-burn of toxic gases
- Smart control: high-temp sensor + microcontroller hold ~800 °C
- Activated carbon + ceramic filters cut smoke and odour
- Gas sensors give alerts, ventilation or auto shut-off
- Handles 4–5 pads per cycle

How it addresses the problem → complete burning, cleaner exhaust, safe use, low-cost school deployment

Dual-chamber + sensing
Re-burns gases and monitors them in real time

Child-safe design
Auto-lock door, insulated body, emergency stop

Low-cost & easy upkeep
Standard power supply, removable ash tray, cleaning alerts

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Technical Approach

Process flow

Load 4–5 pads › Door auto-locks › Primary chamber ~800 °C › Secondary chamber re-burn › Carbon + ceramic filter › Gas sensor check › Clean exhaust

Unsafe gas / over-temperature detected → alert + ventilation or automatic shutdown

Technologies Used

- Arduino Uno microcontroller (C/C++)
- High-temperature sensor + heating coil
- CO/VOC gas sensor, ventilation fan
- Metal heat-resistant dual chambers
- Activated carbon + ceramic filters
- Auto-lock door, buzzer/LED indicators

Links & AI Usage

- GitHub/GitLab: [add link]
- Demo video: [add link]
- Working prototype: dual-chamber unit with sensors
- AI tools: [state extent, e.g. used for research/drafting only]

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Impact Potential

4–5
pads disposed per cycle

~800 °C
controlled complete burning

2-stage
re-burn + filtration cuts smoke & odour

Target Audience Impact

- Girls get safe, private, dignified disposal
- Less stigma and fewer missed school days
- Cleaner air for students and staff

Benefits

- Social: health, dignity, gender equality
- Economic: low cost, fewer toilet blockages
- Environmental: less open burning and pollution

Stakeholders

- Students and teachers
- School administration
- Education & health departments
- NGOs, local bodies, manufacturers

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Feasibility and Risk Analysis

Feasibility

- Prototype in ~3–7 days; installation 1–2 days per school
- Low-cost, off-the-shelf parts; runs on standard electrical supply; needs only basic electronics, Arduino and assembly skills

Challenge / Risk	Strategy to Overcome
Overheating or fire hazard	Temperature sensor with automatic shut-off, insulated body, emergency stop
Toxic smoke, CO/VOC emissions	Secondary re-burn, carbon + ceramic filters, gas sensor alerts
Child injury from hot surfaces	Auto-lock door during operation, warning indicators
Poor maintenance in schools	Removable ash tray, cleaning alert indicator, simple design
Approvals and electrical safety	School administration approval, basic electrical safety compliance

Team Name

Team

Team Composition

[Name 1] Team lead / hardware & Arduino

[Name 2] Research & user interviews

[Name 3] Design, testing & documentation

Edit names and roles to match your team

Why us

- Lived experience of the problem
- Direct student feedback shaped the design

Challenges

- Balancing low cost with safety and emission control
- Reaching ~800 °C safely in a compact unit
- Calibrating sensors and filters

Our Journey

- Started from first-hand experience of smoky school incinerators
- Talked to students; feedback added carbon filter and auto shut-off
- Evolved into a smart, sensor-monitored system

Team Name

References

- WHO/UNICEF Joint Monitoring Programme (JMP) – WASH in Schools and menstrual hygiene data: washdata.org
- UNICEF – Menstrual hygiene and school attendance of girls: unicef.org
- United Nations Sustainable Development Goals 3, 9, 13 (and 5, 6): sdgs.un.org/goals
- Government of India – Menstrual Hygiene Management National Guidelines (2015)
- Solid Waste Management Rules, 2016 (Govt. of India) – sanitary waste handling
- Team field observation and student interviews (primary research)

Please verify each link/title before final submission.